31792 -- Advanced Optimization and Game Theory for Energy Systems

Lecture 4: Market clearing using a cooperative game approach

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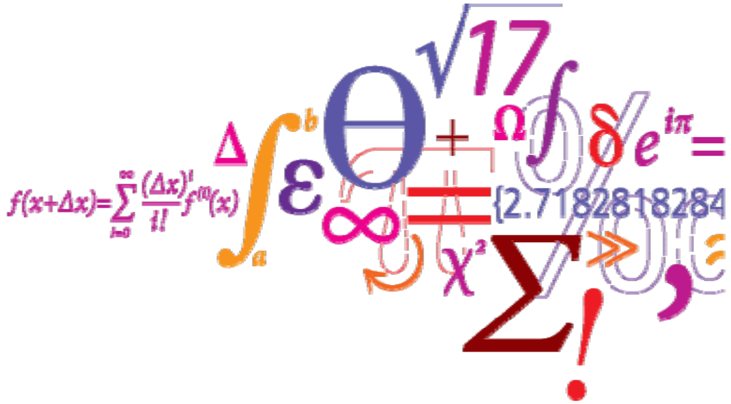

DTU Electrical Engineering
Department of Electrical Engineering

Recap

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- Four desirable properties of market-clearing mechanisms are:
  - Market efficiency:
  - Incentive compatibility:
  - Cost recovery:
  - Revenue adequacy:

Market-clearing	Market efficiency	Incentive compatibility	Cost recovery	Revenue adequacy
LMP				
VCG				

Recap

- Four desirable properties of market-clearing mechanisms are:
  - Market efficiency: maximized social welfare
  - Incentive compatibility: ‘True’ bidding (i.e., capacity and offer price)
  - Cost recovery (individual rationality): non-negative operational profit for all market players
  - Revenue adequacy: no financial deficit for the market operator
    - (balanced budget: no financial deficit *nor excess* for the market operator)

Market-clearing	Market efficiency	Incentive compatibility	Cost recovery	Revenue adequacy
LMP	☹	☹	☺	☺
VCG	☺	☺	☺	☹

No market clearing mechanism can satisfy **all** of these properties simultaneously based on the **Hurwicz Impossibility theorem**!

Objectives of this Lecture

- To review the market-clearing mechanism "Vickrey-Clarke-Groves" (**VCG**), and to implement it on a practical example.
- To introduce the **cooperative game** concept based on the cooperation illustrated in the VCG example, and to understand the differences between a cooperative game and a noncooperative game in terms of their **economic properties**.
- To implement the cooperative game framework in the energy market clearing process, and to discuss the **properties** of different **payoff allocation** methods.

VCG

Consider a market with two generators G1, G2, and two demands D1, D2.

To calculate the payment to G1 under VCG, the market operator clears the market twice: one time including G1, and another time excluding G1.

VCG Payment to G1 = A – B

where term A is calculated based on clearing outcomes when G1 exists in the market:

A = [total utility of demands D1 and D2] –[total cost of generator G2, but not G1]

Similarly, term B is calculated based on market-clearing outcomes when G1 is absent:

B = [total utility of demands D1 and D2] –[total cost of generator G2]

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Recall that VCG ensures “**incentive comaptibility**” and “**market efficiency**” (maximized social welfare)

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A = [total utility of demands D1 and D2] –[total cost of generator G2, but not G1]

A = maximized welfare of {D1, D2, G1, G2} + cost of G1 when in {D1, D2, G1, G2}

Similarly, term B is calculated based on market-clearing outcomes when G1 is absent:

B = [total utility of demands D1 and D2] –[total cost of generator G2]

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VCG

If we Define:

$\mathcal{N}$ = {D1, D2, G1, G2}, $\mathcal{N} \setminus \{G1\}$ = {D1, D2, G2}

$v(\mathcal{N})$ = max. welfare of {D1, D2, G1, G2}, $v(\mathcal{N} \setminus \{G1\})$ = max. welfare of {D1, D2, G2}

$c^{\mathcal{N}}(G1)$ = Total cost of G1 when in {D1, D2, G1, G2}

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VCG Payment to G1 = A – B

$= v(\mathcal{N}) + c^{\mathcal{N}}(G1) - v(\mathcal{N} \setminus \{G1\})$

G1 **net profit** $= v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\})$

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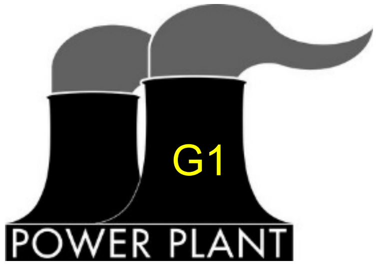
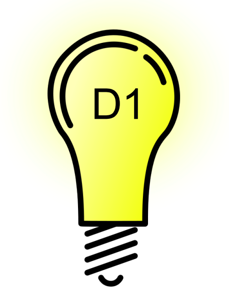
VCG

Let’s try an example!



Capacity: 90 MW
Offer price: $12/MWh

Capacity: 80 MW
Offer price: $20/MWh



Maximum load: 100 MW
Bid price: $40/MWh

Maximum load: 50 MW
Bid price: $35/MWh

Calculate the VCG payments for G1, G2, D1, and D2. What are their net profits?

$$VCG^{G1} = v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\}) - w^{G1,\mathcal{N}*}$$

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Define:

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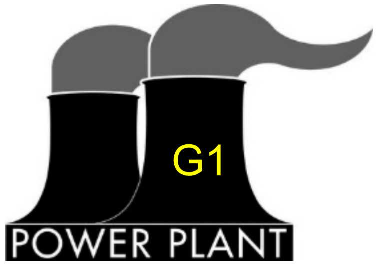

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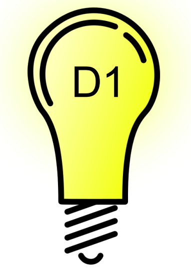

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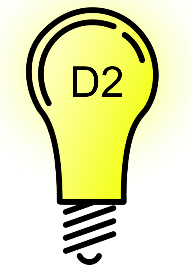

Maximum load: 50 MW
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Calculate the VCG payments for G1, G2, D1, and D2. What are their net profits?

Guide:

1. **Maximize the social welfare** of the scenario with all agents v({G1, G2, D1, D2}), and record the **welfare (utility/cost)** of each agent.
2. Compute the **maximized social welfare** of scenarios **with each agent missing**, e.g., v({~~G1~~, G2, D1, D2}).
3. Follow the steps in the previous slide to compute the VCG **payments** and the **net profits**.

VCG

Let's try an example!

Known:

$$w^{G1} = -12p^{G1}$$
$$w^{G2} = -20p^{G2}$$
$$w^{D1} = 40p^{D1}$$
$$w^{D2} = 35p^{D2}$$

Constraints:

$$0 \leq p^{G1} \leq 90$$
$$0 \leq p^{G2} \leq 80$$
$$0 \leq p^{D1} \leq 100$$
$$0 \leq p^{D2} \leq 50$$
$$p^{G1} + p^{G2} + p^{D1} + p^{D2} = 0$$

$$\boxed{VCG^{G1} = v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\}) - w^{G1,\mathcal{N}*}}$$

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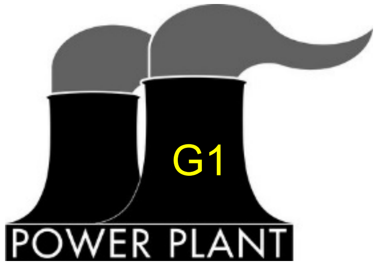

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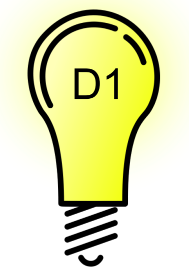

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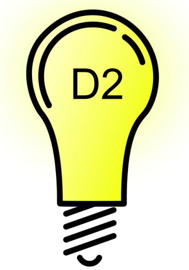

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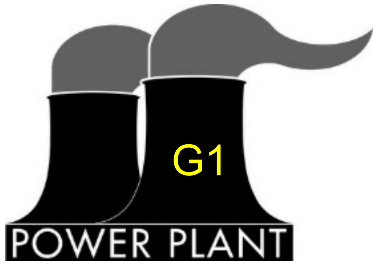

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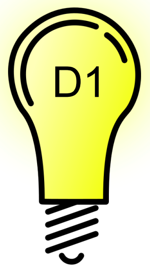

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$\longrightarrow$ Solution: $p^{G1,\mathcal{N}*} = 90 \;\Rightarrow w^{G1,\mathcal{N}*} = -12 * 90 = -\mathbf{1080}$

$$v(\mathcal{N} \setminus \{G1\}) = \max_{p^{Z}, Z \in \mathcal{N}\setminus\{G1\}} \sum_{Z \in \mathcal{N}\setminus\{G1\}} w^{Z} = \max_{p^{G2}, p^{D1}, p^{D2}} (-20p^{G2} + 40p^{D1} + 35p^{D2}) = \mathbf{1600}$$

$$VCG^{G1} = v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\}) - w^{G1,\mathcal{N}*}$$

$$= \mathbf{3470} - \mathbf{1600} - (-\mathbf{1080}) = \mathbf{2950}$$

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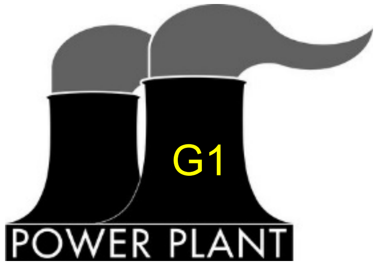

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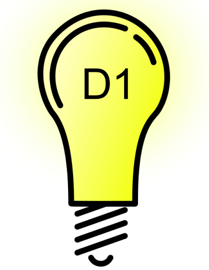

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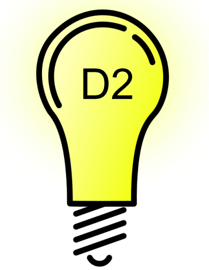

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$$= v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\}) = 1870$$

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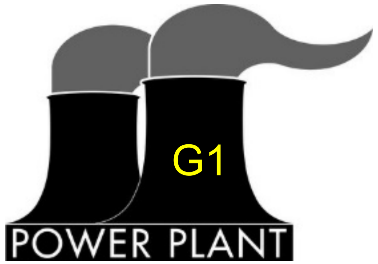

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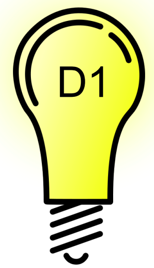

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$v(\mathcal{N})$ and $v(\mathcal{N} \setminus \{G1\})$ can be considered the maximum welfare that can be achieved by the ***coalitions*** {G1, G2, D1, D2} and {G2, D1, D2}, respectively, through cooperation.

The coalition that contains all the players is called the ***grand coalition***. In this example, $\mathcal{N}$ = {D1, D2, G1, G2} is the grand coalition.

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The coalition that contains all the players is called the ***grand coalition***. In this example, $\mathcal{N}$ = {D1, D2, G1, G2} is the grand coalition.

Considering this market as a ***cooperative game***, v is the ***value*** function of this game, and x is the **payoffs** (net profits) for the players.

$$v(\mathcal{N}) = \max_{p^{Z}, Z\in\mathcal{N}} \sum_{Z\in\mathcal{N}} w^Z = \max_{p^{G1}, p^{G2}, p^{D1}, p^{D2}} (-12p^{G1} - 20p^{G2} + 40p^{D1} + 35p^{D2}) = \mathbf{3470}$$

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Cooperative Games

For a more general cooperative game, We define

- $\mathcal{N} = \{1, 2, ..., N\}$ as the grand coalition
- $i \in \mathcal{N}$ as the index for any player in the game, so $i = 1, 2, ..., N$
- $\mathcal{S} \subseteq \mathcal{N}$ as any coalition, which is a subset of the grand coalition

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Given a cooperative game, the definition of the **value function** $v(\mathcal{S})$ of a coalition $\mathcal{S}$ needs to reflect the objectives of the game. A common definition is the maximized total welfare as a result of cooperation among all the players $i \in \mathcal{S}$:

KEY CONCEPT 1

$$v(\mathcal{S}) = \max_{\mathbf{p}_i, \forall i \in \mathcal{S}} \sum_{i \in \mathcal{S}} w_i(\mathbf{p}_i) = \sum_{i \in \mathcal{S}} w_i(\mathbf{p}_i^{\mathcal{S}*})$$

Cooperative Games

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- $\mathcal{N} = \{1, 2, ..., N\}$ as the grand coalition
- $i \in \mathcal{N}$ as the index for any player in the game, so $i = 1, 2, ..., N$
- $\mathcal{S} \subseteq \mathcal{N}$ as any coalition, which is a subset of the grand coalition

Given a cooperative game, the definition of the **value function** $v(\mathcal{S})$ of a coalition $\mathcal{S}$ needs to reflect the objectives of the game. A common definition is the maximized total welfare as a result of cooperation among all the players $i \in \mathcal{S}$:

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KEY CONCEPT 1

decision variables of player i

welfare of player i as a function of the decision variables of player i

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Recall $v(\mathcal{N}) = \max_{p^{Z}, Z \in \mathcal{N}} (-12p^{G1} - 20p^{G2} + 40p^{D1} + 35p^{D2}) = \sum_{Z \in \mathcal{N}} w^{Z}(p^{Z,\mathcal{N}*})$ where $p^{Z,\mathcal{N}*}, \forall Z \in \mathcal{N}$ is the optimal solution

DTU

Cooperative Games

Market-clearing	Market efficiency	Incentive compatibility	Cost recovery	Revenue adequacy
LMP	☹	☹	☺	☺
VCG	☺	☺	☺	☹
Coop	☺	?	Achievable simultaneously	

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Through proper profit allocation mechanisms (discussed later)

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Hurwicz Impossibility theorem!

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Cooperative Games

Clearing mechanism $\xrightarrow{\text{strategies}}$ Equilibrium → **Non-coop framework**

Optimization $\xrightarrow{\text{allocation}}$ Clearing mechanism → **Cooperative framework**

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Optimization $\xrightarrow{\text{allocation}}$ Clearing mechanism → **Cooperative framework**

*VCG can be considered a **non-cooperative** clearing mechanism that leads to a **strategic** equilibrium that matches the optimized market, or a **cooperative profit allocation** mechnism after the market is optimized.

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The redistribution of the **maximized social welfare** is called the **payoff allocation**. We define $\mathbf{x} = \{x_1, \dots, x_N\}$ as the **payoffs** for all the players, which represents the **net profit** of each player after receiving **payment** $\mathbf{r} = \{r_1, \dots, r_N\}$ from the game, where

$$x_i = w_i(\mathbf{p}_i^{\mathcal{N}*}) + r_i$$

KEY CONCEPT 2

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(both terms can be positive or negative)

Recall: $x_{G1}^{VCG} = w^{G1,\mathcal{N}*} + VCG^{G1}$
$= -1080 + 2950 = 1870$

KEY CONCEPT 2

DTU

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Exercise 1

Question:

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Tip: remember from Lecture 3 that the market has a balanced budget if "the market operator has neither financial deficit nor financial excess". In other words, the **total payments** to the market participants should add up to **zero**.

Payoffs for a Cooperative Game

The value function of grand coalition $\mathcal{N}$:

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$$marg_i^{\mathcal{S}} = v(\mathcal{S}) - v(\underline{\underline{\mathcal{S} \setminus \{i\}}}) \qquad \mathcal{S} \setminus \{i\} = \{j | j \in \mathcal{S}, j \neq i\}$$

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RECALL VCG Payment to G1 $VCG^{G1} = \underbrace{v(\mathcal{N}) - v(\mathcal{N} \setminus \{G1\})}_{marg_{G1}^{\mathcal{N}}} - w^{G1,\mathcal{N}*}$

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Therefore, for a general cooperative game, the **payment** based on VCG:

$$\boxed{r_i^{VCG} = marg_i^{\mathcal{N}} - w_i(\mathbf{p}_i^{\mathcal{N}*}) = v(\mathcal{N}) - v(\mathcal{N} \setminus \{i\}) - w_i(\mathbf{p}_i^{\mathcal{N}*})}$$

And the **payoffs** based on VCG: $\boxed{x_i^{VCG} = w_i(\mathbf{p}_i^{\mathcal{N}*}) + r_i^{VCG} = marg_i^{\mathcal{N}}}$

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Payoffs for a Cooperative Game

How do we prove whether VCG satisfies individual rationality?

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Individual Rationality $\Leftrightarrow v(\mathcal{N} \setminus \{i\}) + v(\{i\}) \leq v(\mathcal{N})$

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$\Uparrow$

Superadditivity: $v(\mathcal{S}) + v(\mathcal{T}) \leq v(\mathcal{S} \cup \mathcal{T}), \forall \mathcal{S}, \mathcal{T} \subseteq \mathcal{N}, \mathcal{S} \cap \mathcal{T} = \emptyset$

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What does superadditivity mean in practice?

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What does superadditivity mean in practice?

Known:

$w^{G1} = -12p^{G1}$

$w^{G2} = -20p^{G2}$

$w^{D1} = 40p^{D1}$

$w^{D2} = 35p^{D2}$

Constraints:

$0 \leq p^{G1} \leq 90$

$0 \leq p^{G2} \leq 80$

$0 \leq p^{D1} \leq 100$

$0 \leq p^{D2} \leq 50$

$p^{G1} + p^{G2} + p^{D1} + p^{D2} = 0$

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What does superadditivity mean in practice?

If we define $\mathcal{S} = \{G1, D1\},\ \mathcal{T} = \{G2, D2\},\ \mathcal{N} = \{G1, G2, D1, D2\}$

$$v(\mathcal{S}) = \sum_{Z \in \mathcal{S}} w^Z(p^{G1,\mathcal{S}*} = p^{D1,\mathcal{S}*} = 90) = 2520$$

$$v(\mathcal{T}) = \sum_{Z \in \mathcal{T}} w^Z(p^{G2,\mathcal{T}*} = p^{D2,\mathcal{T}*} = 50) = 750$$

$$v(\mathcal{N}) = \sum_{Z \in \mathcal{N}} w^Z(p^{G1,\mathcal{N}*} = 90, p^{G2,\mathcal{N}*} = 70, p^{D1,\mathcal{N}*} = 100, p^{D2,\mathcal{N}*} = 50) = 3470$$

Known:

$w^{G1} = -12p^{G1}$

$w^{G2} = -20p^{G2}$

$w^{D1} = 40p^{D1}$

$w^{D2} = 35p^{D2}$

Constraints:

$0 \leq p^{G1} \leq 90$

$0 \leq p^{G2} \leq 80$

$0 \leq p^{D1} \leq 100$

$0 \leq p^{D2} \leq 50$

$p^{G1} + p^{G2} + p^{D1} + p^{D2} = 0$

Payoffs for a Cooperative Game

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$$\Rightarrow \boxed{v(\mathcal{S}) + v(\mathcal{T}) = 3270 < v(\mathcal{N}) = v(\mathcal{S} \cup \mathcal{T})}$$

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Two disjoint coalitions cooperating can create a value that is at least as high as the sum of their individual values.

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The value function defined as the maximized energy market's social welfare is **superadditive**

$$v(\mathcal{S}) = \max_{p_i, \forall i \in \mathcal{S}} \sum_{i \in \mathcal{S}} w_i(p_i)$$

Subject to:

- Production limits of generators
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Payoffs for a Cooperative Game

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since $\mathbf{p}_i^{\mathcal{S}*}, \mathbf{p}_i^{\mathcal{T}*}$ are the solutions for $v(\mathcal{S}), v(\mathcal{T})$, $\mathbf{p}_i^{\mathcal{S} \cup \mathcal{T}}$ satifies the constraints of all agents in $\mathcal{S}$ and $\mathcal{T}$.

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Counter examples?

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How do we modify VCG into a budget balancing imputation

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$\frac{1}{N}$

$$-v(\mathcal{N} \setminus \{i\}) + [v(\mathcal{N} \setminus \{i\}) - v(\mathcal{N} \setminus \{i\} \setminus \{j\})] + \ldots + [v(\{k\}) - v(\emptyset)]$$

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Shapley value[1]: $\phi_i = \frac{1}{N} \sum_{\mathcal{S}\subseteq\mathcal{N}, i\in\mathcal{S}} \binom{N-1}{|\mathcal{S}|-1}^{-1} [v(\mathcal{S}) - v(\mathcal{S}\setminus\{i\})]$

(Average marginal contribution of a player to all the coalitions that contain this player[2])

[1] L. Shapley, "A Value for n-person Games," *Contributions to the Theory of Games,* vol. 2, no. 28, pp. 307–317, 1953

[2] J. Castro, D. Gómez, and J. Tejada, “Polynomial calculation of the Shapley value based on sampling,” *Computers and Operations Research*, vol. 36, no. 5, pp. 1726–1730, 2009.

The Shapley Value

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$$= \sum_{\mathcal{S} \subseteq \mathcal{N}, i \in \mathcal{S}} \frac{(|\mathcal{S}|-1)!(N-|\mathcal{S}|)!}{N!} [v(\mathcal{S}) - v(\mathcal{S} \setminus \{i\})]$$

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Given **Superadditivity**: $v(\mathcal{S}) + v(\mathcal{T}) \leq v(\mathcal{S} \cup \mathcal{T}), \forall \mathcal{S}, \mathcal{T} \subseteq \mathcal{N}, \mathcal{S} \cap \mathcal{T} = \emptyset$

The Shapley Value

A **payoff allocation** is called an **imputation** if it satisfies

1) Individual Rationality: ☺ $x_i \geq w_i(\mathbf{p}_i^{\{i\}*})$

2) Balanced Budget: ☺ $\sum_{i \in \mathcal{N}} x_i = v(\mathcal{N})$

Shapley value:

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The Shapley value is an imputation!

$\Uparrow$

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The Shapley value is an imputation!

Try to prove it yourself following similar steps in the VCG proofs

Remember: the Shapley value is the average marginal contribution of a player to all the coalitions that contain this player.

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Remember: the Shapley value is the average marginal contribution of a player to all the coalitions that contain this player.

Additional properties of the Shapley value:

3) Symmetry: if for all $\mathcal{S}$ that contains neither i nor j, $v(\mathcal{S} \cup \{i\}) = v(\mathcal{S} \cup \{j\})$, then $\phi_i = \phi_j$

4) Dummy player: if for all $\mathcal{S}$ that does not contain i, $v(\mathcal{S} \cup \{i\}) - v(\mathcal{S}) = v(\{i\})$, then $\phi_i = v(\{i\})$

5) Linearity: if two games of values v and u are played together, then $\phi_i(v+u) = \phi_i(v) + \phi_i(u)$

Fairness properties[3]!

[3] R. van den Brink, "An axiomatization of the Shapley value using a fairness property", *International Journal of Game Theory*, vol. 30, no. 3, 309–319, 2002.

Computation of the Shapley Value

Shapley value:

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Example of a 3-player game

	{1}	{2}	{3}	{1, 2}	{1, 3}	{2, 3}	{1, 2, 3}
v	0	0	0	0	5	9	10

Calculate the Shapley value for all the players.

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$v(\emptyset) = 0$

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$marg_1$							
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How many coalitions are there for an N-player game?

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DTU

How many coalitions are there for an N-player game?

Computation of the Shapley Value

2^N (Intractable) → **Sampling** for approximating the Shaley value[2]

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Stability of a Cooperative Game

Notice $\phi_2 + \phi_3 = \frac{19+34}{6} = \frac{53}{6} < 9 = v(\{2,3\})$

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We define $e_{\mathcal{S}}^{\mathbf{x}}$ as the **excess** of coalition $\mathcal{S}$ given payoff allocation $\mathbf{x} = \{x_1, ..., x_N\}$:

KEY CONCEPT 3

$$e_{\mathcal{S}}^{\mathbf{x}} = v(\mathcal{S}) - \sum_{i \in \mathcal{S}} x_i$$

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$$e_{\mathcal{S}}^{\mathbf{x}} \leq 0, \ \forall \mathcal{S} \subseteq \mathcal{N}$$

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$\Longrightarrow$ The Shapley value is not necessarily in the **core**

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In fact, the **core** of a cooperative game can be **empty,** meaning not every game has stabilizing payoffs.

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Is there a payoff allocation that is guaranteed to be in the core **if** the core is **non-empty**?

We define $e^{\mathbf{x}}_{\mathcal{S}}$ as the **excess** of coalition $\mathcal{S}$ given payoff allocation $\mathbf{x} = \{x_1, ..., x_N\}$:

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YES! The **nucleolus** has the lexicographically minimum excesses[4][5] among all imputations.

[4] Schmeidler, David. "The Nucleolus of a Characteristic Function Game." *SIAM Journal on Applied Mathematics*, vol. 17, no. 6, 1969, pp. 1163–1170.

[5] E. Baeyens, E. Y. Bitar, P. P. Khargonekar and K. Poolla, "Coalitional Aggregation of Wind Power," in *IEEE Transactions on Power Systems*, vol. 28, no. 4, pp. 3774-3784, Nov. 2013

References

- Seminal work on the Shapley value

[1] L. Shapley, "A Value for n-person Games," *Contributions to the Theory of Games,* vol. 2, no. 28, pp. 307–317, 1953

- Alternative expression of the Shapley value based on marginal contributions to all coalitions, and the approximation of the Shapley value using sampling methods

[2] J. Castro, D. Gómez, and J. Tejada, "Polynomial calculation of the Shapley value based on sampling," *Computers and Operations Research*, vol. 36, no. 5, pp. 1726–1730, 2009.

- Fairness properties of the Shapley value

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- The stability of a cooperative game and the nucleolus payoff allocation

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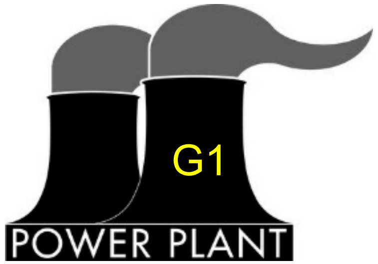
Take-Aways

- VCG is a market clearing mechanism that ensures that the strategic market equilibrium yields an **optimized social welfare**. It can also be considered as a payoff allocation of a **cooperative game** that equals the **marginal contribution** of a player to the **grand coalition**.
- Assuming players provide truthful information, a cooperative game first maximizes the social welfare (guaranteed **market efficiency**), then allocates the payoffs to the participants. This **payoff allocation** is said to be an **imputation** if it satisfies **individual rationality** and **balanced budget**.
- The **Shapley value** is an imputation of a **superadditive** cooperative game. It is calculated by averaging a player's marginal contributions to all possible (**2^N**) **coalitions** in the game.
- The **core** of a cooperative game is the set of all **stabilizing** payoff allocations, i.e., payoff allocations that guarantee non-positive **excesses** for all coalitions.
  - The core can be empty.
  - The Shapley value is not guaranteed to be in the core.
  - If the core is non-empty, the **nucleolus** is guaranteed to be in the core.

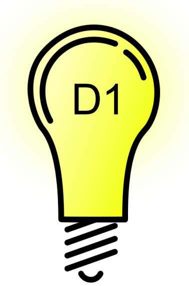
Exercise 2

Recall example



Capacity: 90 MW
Offer price: $12/MWh

Capacity: 80 MW
Offer price: $20/MWh



Maximum load: 100 MW
Bid price: $40/MWh

Maximum load: 50 MW
Bid price: $35/MWh

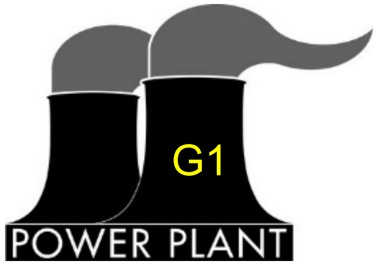
Question:

- Assuming the generators and loads are participants in a cooperative game, what are the payments to each participant based on the Shapley value? Compare the results with VCG.
- Are the resulting payoffs from the Shapley value stabilizing (i.e., is the Shapley value in the core)?

Note: Make sure to distinguish between the payoffs and the payments

Exercise 3

Recall example



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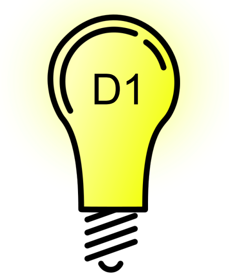
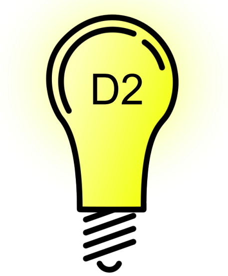


Maximum load: 100 MW
Bid price: $40/MWh



Maximum load: 50 MW
Bid price: $35/MWh

Question:

- Does this game have a non-empty core? If so, how do you come up with a stabilizing payoff allocation? Please write a program for this purpose.

Guide:

1. Compute the values of the coalitions (you would have done this already to compute the Shapley value).
2. Construct a linear problem with the 4 payoffs as variables, and the highest excess as the 5th variable.
3. Write the highest excess as constraints for all the coalitions.
4. Minimize the highest excess. Is it negative or positive? What does it mean?

(This is the 1st step towards computing the nucleolus)

Thanks for your attention!

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